

What to Learn from Fama & French (1993)

"Common Risk Factors in the Returns on Stocks and Bonds"

A plain-language guide to the key ideas, methods, and practical takeaways
for our ML Factor Investing project on the S&P 500.

Prepared for the ML Factor Investing Team

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1. What the Paper Is About

Fama and French (1993) is one of the most cited papers in finance. It asks a simple question: what common risk factors explain why some stocks earn higher average returns than others? Before this paper, the standard model (the CAPM) said that only one thing mattered: how sensitive a stock is to the overall market (its beta). Fama and French showed that beta alone does a poor job, and that two additional factors, related to company size and book-to-market equity, do much better.

The paper identifies five risk factors in total: three for stocks and two for bonds. It shows that these five factors, used together, capture most of the common variation in stock and bond returns and explain the cross-section of average returns (meaning: why some portfolios earn more than others over time).

This paper is the intellectual foundation for factor-based investing. The features we use in our ML model (size, value, momentum, etc.) descend directly from the research program this paper launched.

2. The Five Risk Factors Explained Simply

2.1 The Three Stock-Market Factors

Market Factor (RM - RF). This is the return on the overall stock market minus the risk-free rate (one-month Treasury bill). It captures the basic reward for holding stocks instead of safe bonds. Every stock is exposed to this factor. In CAPM, this was the only factor that mattered.

Size Factor (SMB = Small Minus Big). Each month, compute the average return on small-company stock portfolios minus the average return on big-company stock portfolios. Historically, small stocks earn higher average returns than big stocks. SMB captures this size premium. A stock with a high loading (slope) on SMB behaves like a small stock; a stock with a low or negative loading behaves like a big stock.

Value Factor (HML = High Minus Low). Each month, compute the average return on high book-to-market stocks minus the average return on low book-to-market stocks. Book-to-market equity (BE/ME) is the ratio of a company's book value to its market value. High BE/ME stocks are called value stocks (cheap relative to their books); low BE/ME stocks are called growth stocks (expensive relative to their books). Historically, value stocks earn higher average returns. HML captures this value premium.

2.2 The Two Bond-Market Factors

Term Factor (TERM). The return on long-term government bonds minus the one-month T-bill rate. This captures the risk premium for holding long-duration bonds, which are sensitive to unexpected changes in interest rates. When interest rates shift, long-term bonds move more than short-term ones.

Default Factor (DEF). The return on a portfolio of long-term corporate bonds minus the return on long-term government bonds. This captures the risk premium for bearing default risk: the chance that a company cannot pay back its debt. When economic conditions worsen, default risk rises and corporate bonds underperform government bonds.

For our project: the three stock factors (market, size, value) inform our ML features. The two bond factors (TERM, DEF) are closely related to the macro variables our regime model uses (yield curve slope and credit spreads).

3. How Fama & French Build Their Factors

Understanding the construction method matters because it teaches us about careful data handling and avoiding look-ahead bias.

3.1 The Double Sort

Every June, Fama and French sort all NYSE stocks into two groups by size (market capitalization measured at end of June): small (below NYSE median) and big (above). Independently, they sort all NYSE stocks into three groups by book-to-market equity: low (bottom 30%), medium (middle 40%), and high (top 30%). The breakpoints come only from NYSE stocks, but all NYSE, Amex, and NASDAQ stocks are then placed into the resulting six portfolios (2 size groups x 3 BE/ME groups).

3.2 Why NYSE Breakpoints Matter

Using NYSE breakpoints prevents tiny, illiquid stocks (many on Amex and NASDAQ) from dominating the sort. Without this, the 'small' group would contain thousands of micro-cap stocks that are hard to trade in practice. This is an early lesson in making academic research implementable. For our project, the S&P 500 universe avoids this specific issue, but the principle applies: when ranking stocks within sectors, be aware that sector sizes vary enormously (Tech has 70+ stocks, Energy has 22).

3.3 The Timing Rules (Critical for Our Project)

Portfolios are formed in June of year t . Book equity (BE) comes from the fiscal year ending in calendar year $t-1$. Market equity (ME) for the BE/ME ratio is measured at end of December of $t-1$. Returns are then calculated from July of year t to June of year $t+1$.

Why the gap? To ensure the accounting data was publicly available before it enters the model. A company's fiscal year ending December 2023 might not file its annual report until March or April 2024. Fama and French wait until June to be safe. This is a 6-month lag.

This is the origin of the reporting lag discipline in our project. Our framework enforces a 45-day lag on fundamentals (tighter than 6 months because modern SEC filings are faster). The lesson: never use accounting data in your model before it was publicly available. Violating this creates look-ahead bias and inflates your backtest results.

4. The Key Findings

4.1 The Three Stock Factors Explain Stock Returns Well

When Fama and French regress the returns of 25 stock portfolios (sorted by size and BE/ME) on RM-RF, SMB, and HML, the typical R-squared values are above 0.93. The regression intercepts (alphas) are close to zero, meaning the three factors account for nearly all the differences in

average returns across these portfolios. By contrast, the CAPM (market factor alone) leaves large unexplained patterns: small stocks and high-BE/ME stocks have returns that are systematically too high relative to what their market beta predicts.

4.2 SMB Slopes Track Size, HML Slopes Track Value

The SMB slopes decrease monotonically from the smallest to the biggest size quintile in every BE/ME group. The HML slopes increase monotonically from the lowest to the highest BE/ME quintile in every size group. This is exactly what you would expect if SMB and HML are capturing genuine common risk factors related to size and value.

4.3 TERM and DEF Capture Bond Returns (and Show Up in Stocks)

The two bond factors, TERM and DEF, explain almost all the common variation in government and corporate bond returns (R-squared near 0.97 for high-grade corporates). They also show up in stock returns: stock portfolios have TERM slopes around 0.8-1.0, similar to long-term bonds. This means stocks are sensitive to the same interest rate shocks that move long-term bonds. However, the average premiums for TERM and DEF are tiny (0.06% and 0.02% per month), so these factors explain almost none of the cross-sectional differences in average stock returns.

Key insight: TERM and DEF explain when returns are high or low for everyone (time-series), but not which stocks earn more than others (cross-section). This directly validates our project's two-component architecture.

5. Cross-Sectional vs. Time-Series Variation

This is the single most important conceptual distinction in the paper for our project. Make sure you understand it clearly.

5.1 What Cross-Sectional Variation Means

Cross-sectional variation is about differences across stocks at a given point in time. Why does Portfolio A earn 1.05% per month on average while Portfolio B earns 0.32%? The answer is that A loads more heavily on SMB and HML. The stock-specific factors (size, value) explain which stocks earn more.

5.2 What Time-Series Variation Means

Time-series variation is about changes over time that affect all stocks similarly. Why were stock returns high in 1995 and low in 2008? The answer involves macro conditions: interest rate shocks (TERM), default risk (DEF), and overall market risk (RM-RF). These factors explain when returns are high or low for everyone.

5.3 Why This Split Matters for Our Architecture

Our ML prediction model ranks stocks cross-sectionally: given features, which stocks will outperform and which will underperform next month? The features that help with this are firm-level characteristics (size, value, momentum, profitability). These are the descendants of SMB and HML.

Our regime model uses macro variables (VIX, yield curve slope, credit spreads) to detect market conditions. These are the descendants of TERM and DEF. They tell us about the overall risk environment, not about individual stock selection.

Fama and French proved empirically that these two types of information serve different purposes. That is the theoretical foundation for why our project separates them into two distinct components: the ML model (cross-sectional ranking) and the regime overlay (time-series risk management).

6. The Reporting Lag Discipline

Fama and French are meticulous about one rule: never use data before it was publicly available.

Data Item	Measured At	First Used In Portfolio
Market equity for size sort	End of June, year t	July of year t
Book equity (BE)	Fiscal year ending in year t-1	July of year t (6-month lag)
Market equity for BE/ME	End of December, year t-1	July of year t

Portfolio returns	July year t to June year t+1	N/A (these are outcomes)
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Our project rule: apply a 45-day reporting lag on all fundamental data. A Q1 report (quarter ending March 31) is not usable until mid-May at the earliest. This must be enforced in the data pipeline code (data_loader.py) so that no downstream code can accidentally violate it.

7. What This Means for Our ML Features

The paper tells us which characteristics carry genuine information about expected stock returns. Here is how the paper's findings translate into features for our XGBoost model.

FF93 Finding	Feature for Our Model	How to Compute
Size (ME) predicts returns; small stocks earn more	Market cap rank (sector-relative)	Within each sector, rank stocks by market cap each month
Book-to-market (BE/ME) predicts returns; value stocks earn more	Book-to-market ratio (sector-relative)	Book equity / market equity, ranked within sector
Earnings/price (E/P) shows U-shaped returns	Earnings yield (sector-relative)	Earnings / price, ranked within sector
Dividend/price (D/P) shows U-shaped returns	Dividend yield (sector-relative)	Dividends / price, ranked within sector
Market beta alone does not explain returns well	Do not rely on beta as a solo feature	Can include but expect low marginal importance

All features must be computed as sector-relative ranks or z-scores (Layer 1 of our feature stack). This removes sector imbalance and makes features cross-sector comparable. A utility stock ranked 0.9 is in the top 10% of utilities; a tech stock ranked 0.9 is in the top 10% of tech.

Important: Fama and French use raw characteristics. Our ML model consumes the same underlying information but in sector-relative form, which the paper's own findings suggest is the right approach for a sector-neutral long-short strategy.

8. What This Means for Our Regime Model

The paper's bond-market factors, TERM and DEF, are the academic versions of the macro variables our GMM regime model uses. Here is the mapping:

FF93 Factor	Definition	Our Regime Feature
TERM	Long-term govt bond return minus T-bill rate	Yield curve slope (10Y - 2Y Treasury spread)
DEF	Corporate bond return minus govt bond return	Credit spread (BAA - AAA or HY - IG spread)
(Implicit in market factor)	Market volatility and crisis conditions	VIX level, realized market volatility

The paper shows that TERM and DEF capture strong common variation in returns over time, but their average premiums are close to zero, meaning they do not help rank stocks cross-sectionally.

This is exactly why we keep these variables in the regime model (time-series risk management) and do not feed the regime label into the return prediction model.

The paper also shows that expected TERM and DEF premiums vary with business conditions: the term premium is positive near recessions and negative near expansions; the default premium is high when the economy is weak. This is the academic evidence that macro conditions cluster into regimes, supporting our use of a Gaussian Mixture Model to detect them.

9. What This Means for Portfolio Construction

Fama and French's portfolio construction method has several lessons for our strategy:

- **Sector-neutral construction prevents concentration.** Fama and French construct SMB so that the small and big portfolios have similar book-to-market characteristics, and HML so that the high and low BE/ME portfolios have similar size. This orthogonalization ensures each factor captures only its intended effect. Similarly, our strategy constructs portfolios that are sector-neutral: we go long the top stocks within each sector and short the bottom stocks within each sector, ensuring we are not accidentally betting on sector exposures.
- **Value-weighted components reduce variance.** The paper uses value-weighted returns within each portfolio. For our strategy, this means that when we allocate within each sector leg, weighting toward larger, more liquid names (or at least not equal-weighting tiny positions) reduces both variance and transaction costs.
- **Annual rebalancing vs. monthly.** Fama and French rebalance their factor portfolios annually (every June). Our strategy rebalances monthly because the ML model generates fresh predictions each month. This means turnover management is more critical for us. The paper's low-frequency rebalancing is a reminder that not all signal needs to be acted on immediately.

10. Using FF3 as a Benchmark for Our Strategy

One of the most practical applications of this paper is as a performance evaluation tool. After our backtest is complete, we should regress our strategy's monthly returns on the three Fama-French factors:

$$R_{\text{strategy}}(t) - RF(t) = \alpha + b \times (RM(t) - RF(t)) + s \times SMB(t) + h \times HML(t) + \text{error}(t)$$

What the results tell us:

- **Alpha (intercept):** This is the average monthly return our strategy earns after accounting for its exposure to the market, size, and value factors. If alpha is significantly positive, our ML model is capturing something beyond known risk premia. This is the most important number in the regression.
- **Factor loadings (b, s, h):** These tell us what risks our strategy is exposed to. If s is large and positive, our strategy behaves like small stocks. If h is large and positive, it behaves like value stocks. A well-designed market-neutral strategy should have b close to zero (minimal market exposure) and ideally modest s and h (meaning returns do not simply come from known factor tilts).
- **R-squared:** How much of our strategy's return variation is explained by the three factors. A low R-squared means our strategy has significant idiosyncratic returns (good for a hedge fund). A high R-squared means we are mostly replicating known factor exposures.

The Fama-French factor return series are freely available from Kenneth French's website (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). Download the monthly factors file and merge it with our backtest output by date.

11. Common Misunderstandings to Avoid

Misunderstanding: "We should build SMB and HML as trading strategies."

Correct view: No. SMB and HML are factor-mimicking portfolios used for regression analysis and benchmarking. Our ML model consumes the raw characteristics (size, B/M ratio) as features and learns the mapping itself. We only need the factor return series to evaluate our strategy after the fact.

Misunderstanding: "Market beta is useless."

Correct view: The paper says beta does not explain the cross-section of average returns well, not that beta is irrelevant. Beta still captures market exposure. For a long-short portfolio, we want net beta near zero, which is a portfolio construction constraint, not a feature selection decision.

Misunderstanding: "TERM and DEF should be features in our ML model."

Correct view: As continuous variables (yield curve slope, credit spread), they can be useful features. But the paper shows their average premiums are near zero cross-sectionally. Their main value is in the regime model. Do not confuse the raw macro variables (which can help the ML model see interactions) with the discrete regime label (which should stay outside the prediction model).

Misunderstanding: "The three-factor model is the final word."

Correct view: Fama and French themselves later extended it to five factors (2015) by adding profitability and investment. Momentum (Carhart, 1997) is another well-documented factor. Our ML model should include features related to all of these: size, value, momentum, profitability, and investment. The 1993 paper is the starting point, not the ceiling.

Misunderstanding: "Sorting stocks into quintiles is how we should construct our portfolio."

Correct view: The double sort is a research methodology for isolating factor effects. Our portfolio construction uses sector-neutral ranking driven by ML predictions. We rank stocks by predicted return within each sector, then go long the top k and short the bottom k per sector.

12. Summary: The Complete Checklist

Here is everything you should take from this paper, organized by who on the team it matters most for.

For Person A (Data & Infrastructure)

- Enforce reporting lag on all fundamental data (45-day minimum). Fama-French use a 6-month lag; modern filings allow tighter windows, but never zero.
- Include book-to-market equity, earnings/price, and dividend/price in the data pipeline. These are the characteristics the paper validates as return predictors.
- Download the Fama-French factor return series from Kenneth French's data library and include them in the backtest output for post-hoc performance attribution.
- When computing features at each rebalance date, use only data that was publicly available on that date. This applies to all accounting variables.

For Person B (ML Prediction Model)

- Use size (market cap), book-to-market, earnings yield, and dividend yield as core features. These are the paper's validated return predictors.
- Compute all features as sector-relative ranks. This removes sector imbalance and makes the model's learning objective sector-neutral.
- Do not rely on market beta as a primary feature. The paper shows it has little cross-sectional explanatory power.
- Plan to expand beyond FF93: add momentum, profitability, and investment features. These come from later research in the same tradition.
- After the backtest, regress strategy returns on the three FF factors. A significant positive alpha means the model is adding value beyond known risk premia.

For Person C (Regime Model)

- Use yield curve slope and credit spreads as core regime features. These are the real-world equivalents of the paper's TERM and DEF factors.
- Understand why TERM and DEF are time-series factors, not cross-sectional ones. They explain when markets are risky, not which stocks win. This justifies keeping them in the regime model, separate from the prediction model.
- The paper shows TERM and DEF premiums vary with business conditions (positive near troughs, negative near peaks). This is empirical evidence that macro conditions cluster into regimes, supporting the GMM approach.
- Do not feed the regime label into the ML prediction model. The paper's finding that TERM and DEF have negligible cross-sectional explanatory power is the reason.

For the Whole Team

- Understand the cross-section vs. time-series distinction. It is the intellectual foundation for our two-component architecture (ML model + regime overlay).
- Never use data before it was publicly available. This is the single most important rule in financial backtesting, and Fama-French is where the discipline originates.
- The three-factor model is a benchmark, not a competitor. We are trying to beat it, not replicate it. If our strategy alpha (after FF3 regression) is significantly positive, we have succeeded.
- Read the Fama-French (2015) five-factor paper as a follow-up. It adds profitability (RMW) and investment (CMA) factors that should also be in our feature set.

Reference: Fama, E.F. and French, K.R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3-56.